Think

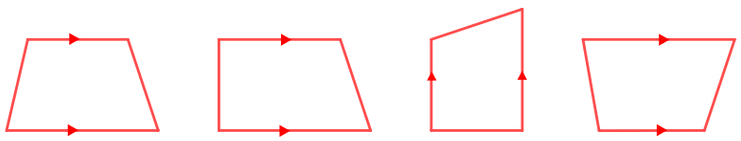
Is it possible to construct a quadrilateral $PQRS$ with $PQ = 5$ cm, $QR = 3$ cm, $RS = 6$ cm, $PS = 7$ cm and $PR = 10$ cm. If not, why?

5.11 Construction of Trapeziums

In the first term, we have learnt how to construct the quadrilaterals. To draw a quadrilateral, how many measurements do you need? 5 measurements. Isn't it? Let us see the special quadrilaterals which need less than 5 measurements. Based on the nature of sides and angles of a quadrilateral, it gets special names like trapezium, parallelogram, rhombus, rectangle, square and kite.

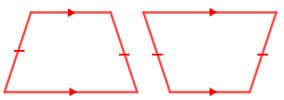
Now, you will learn how to construct trapeziums.

Trapezium is a quadrilateral in which a pair of opposite sides are parallel. To construct a trapezium, draw one of the parallel sides as a base and on that base construct a triangle with the 2 more measurements. Now, through the vertex of that triangle, construct the parallel line opposite to the base so that the triangle lies between the parallel sides. As the fourth vertex lies on this parallel line, mark it with the remaining measure. Hence, we need four independent measures to construct a trapezium. The given shapes are examples of trapeziums.



Note: The arrow marks in the above shapes represent parallel sides.

If the non-parallel sides of a trapezium are equal in length and form equal angles at one of its bases, then it is called an **isosceles trapezium**.



Try these

1. The area of the trapezium is _____.
2. The distance between the parallel sides of a trapezium is called as _____.
3. If the height and parallel sides of a trapezium are 5*cm*, 7*cm* and 5*cm* respectively, then its area is _____.
4. In an isosceles trapezium, the non-parallel sides are _____ in length.
5. To construct a trapezium, _____ measurements are enough.
6. If the area and sum of the parallel sides are 60 cm^2 and 12 cm , its height is _____.

Let us construct a trapezium with the given measurements

1. Three sides and one diagonal.
2. Three sides and one angle.
3. Two sides and two angles.
4. Four sides.

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